

Linear regression

$$\mathbf{w}^\top \mathbf{x} + b$$

Part I

+ output
link

Logistic/softmax

$$\text{softmax}(\mathbf{W}\mathbf{x} + \mathbf{b})$$

Part I

+ hidden
bend

MLP

$$\mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1)$$

Part I

+ spatial sharing

CNN

one kernel,
every position

Part II

+ temporal
sharing

RNN

one update,
every step

Part III

+ learned
similarity

Attention

$$\text{softmax}(\mathbf{q}^\top \mathbf{K}^\top) \mathbf{V}$$

Part IV

+ reuse

Pretrained model

learn once,
adapt often

Part V